

**PsyClick: Clinical Decision Support System for Detecting Single-Session  
Psychomotor Disturbances via EWMA-Based Multivariate Analysis  
of Keystroke and Cursor Dynamics**

by

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**Live Demonstration Script**

FEU Institute of Technology  
July 2026

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*How to read this: text in normal type is spoken aloud. Text in [ gray brackets ] is an action to perform — do it, do not read it. Speak slowly; pause where you see a paragraph break. Estimated time: 15-18 minutes.*

## Opening

*[ Have the app open on the landing screen, signed out. Screen visible to the panel. ]*

Good morning. I'm going to give you a live walkthrough of PsyClick Secure — our hardened clinical decision-support system. Everything you'll see runs entirely on this workstation, offline, with no internet connection, and we're using only synthetic, non-clinical data throughout.

I'll cover five things: how a clinician logs in, how we manage users and roles, the security controls built into the system, our audit trail, and finally a complete assessment session from start to finish.

## Part 1 — User Login

Let's start with signing in. Access to PsyClick is by individual clinician account — there are no shared logins.

*[ Enter the clinician ID and password for Dr. Clinician, and sign in. ]*

When I sign in, the system opens the clinician dashboard, and behind the scenes it records a login event to the audit trail — I'll show you that trail later.

A quick word on how we protect this login. Passwords are never stored in plain text — they're protected with scrypt, a memory-hard hashing algorithm, and we enforce a minimum length of twelve characters. Each session expires automatically when it's idle, and can be revoked at any time.

Now let me show you what happens when someone tries to break in.

*[ Sign out. Then deliberately type a wrong password a few times on one account. ]*

Every failed attempt is rejected and logged. After five failed attempts on a single account, that account is locked for fifteen minutes. And independently, if the system sees a burst of failures coming from one source, it throttles that source for about two minutes and writes a security alert.

*[ Continue the failed attempts until you see the "too many attempts" message. ]*

There it is — the intrusion monitor has kicked in. This is our first line of defense against password guessing and credential-stuffing attacks, and it's working exactly as designed.

## Part 2 — Role Management

Next, let's look at how we manage users and their roles. For this I'll sign in as an administrator.

*[ Sign in as Dr. Admin. Open the Security Center. ]*

PsyClick has three roles: administrator, clinician, and auditor. From the Security Center, an administrator can create accounts and assign each of these roles.

*[ Create a new clinician account. Then create an auditor account, selecting the role from the dropdown. ]*

I've just created a clinician and an auditor. Here's the account list, showing each user, their role, and their status. An administrator can also manage privileges — for example, deactivating or locking an account.

Now, the important part — this isn't just labels. The roles are actually enforced. Let me prove it.

*[ Sign out. Sign in as the clinician. Try to open the account-management / admin area. ]*

As a clinician, when I try to reach the administrator area, the system denies it — insufficient permission. Clinicians are also scoped to their own records, so one clinician cannot open another clinician's sessions.

*[ Sign out. Sign in as the auditor. Open the audit log; then try to open account management. ]*

And as an auditor, I can review the audit log — that's the auditor's job — but I'm blocked from managing accounts. This is least privilege: each role can do only what it needs to, and every denial is itself recorded as a security event.

## Part 3 — Security Controls

Now let me highlight a few of the security controls built into the workflow, starting with consent.

*[ As the clinician, create a session for the synthetic client DEMO-001, but do not record consent yet. Try to start the assessment. ]*

Notice what just happened. I tried to start capturing data before recording the participant's consent, and the system stopped me — it says recorded informed consent is required. This check runs on the server, not just in the interface, so capture simply cannot begin until consent exists. We call this fail-closed: if in doubt, it blocks.

*[ Record consent, then show that the assessment can now proceed. ]*

Once consent is recorded, the session proceeds normally.

Second, data protection. Let me create a backup.

*[ Sign in as Dr. Admin if needed. Create an encrypted backup, then run backup verification. ]*

Our backups and exports are encrypted using Windows DPAPI, and every backup is stamped with a SHA-256 integrity digest. When I verify the backup, the system re-checks that digest and validates the copy in an isolated location — so a corrupted or tampered backup is caught before anyone relies on it. And again, both the backup and the verification are written to the audit log.

## Part 4 — Audit Logs

That brings us to the audit trail, which ties everything together.

*[ Open the Audit page. ]*

This is our accountability record. Every meaningful action is here — logins, permission denials, consent, record access, exports, configuration changes, and backups — each stamped with who did it and when. You can see the events from this very demonstration listed in order.

But here's what makes it trustworthy: the log is hash-chained. Each entry carries a fingerprint of the entry before it, so if anyone edits or deletes even a single row, the chain breaks. Let me verify it.

*[ Run the audit-chain verification. ]*

The verification confirms the chain is intact — no entries have been altered. That makes our audit log tamper-evident, which is essential for accountability in a clinical setting.

## **Part 5 — Core System Functionality**

Finally, let me walk you through a complete assessment session, which is what PsyClick is actually for.

*[ As the clinician, select client DEMO-001 with consent recorded. Begin the session. ]*

PsyClick measures how a person types and moves the mouse — the timing and rhythm of their movements — to detect subtle psychomotor changes. We start by calibrating the keyboard and the mouse so the measurements are consistent across devices.

*[ Run keyboard calibration, then mouse calibration. ]*

Next, the participant completes two standard screening questionnaires — the PHQ-9 and the GAD-7 — followed by a few short emotional-task prompts, where we capture keystroke and cursor dynamics as they type.

*[ Administer PHQ-9 and GAD-7, then run the emotional-task prompts with short typed responses. ]*

Now I'll let the system finalize and compute the analysis. I won't close the app while it's processing.

*[ Let the session finalize. Open the dashboard / report. ]*

And here are the results. This is the T-squared score — a single multivariate measure of how far this session deviates from the person's own baseline — shown against its threshold. These two indices, PSI and PAI, break that score down: they

tell the clinician which behaviors drove it, whether the pattern looks like slowing or agitation.

The overall signal is summarized as green, amber, or red, and — this is the key point — every flag comes with a plain-language rationale explaining why. PsyClick is a decision-support tool, not a diagnostic one. It amplifies a behavioral signal for the clinician; the clinician always combines it with their own professional judgment.

*[ Export the session report to the approved protected destination. ]*

Finally, I'll export the session report to a protected destination — and, as you'd expect by now, that export is encrypted and recorded in the audit log.

### Closing

So to recap: individual, hardened logins with lockout and intrusion monitoring; real role-based access control; consent enforced fail-closed; encrypted, verifiable backups; a tamper-evident audit trail; and a complete, explainable, non-diagnostic assessment — all running securely and entirely offline on a single workstation.

That concludes the demonstration. We're happy to take your questions.

### If Something Goes Wrong (do not read)

- *"API not available": restart the app; confirm the backend is on 127.0.0.1:5101; no internet is needed.*
- *Locked out or throttled mid-demo: wait for the cooldown, or switch to another pre-created account. Do not disable the monitor.*
- *A step stalls: move to the next part and return later — the audit log preserves what was completed.*
- *Use synthetic data only. Never show real credentials or patient information on screen.*